UPVAAS-Twin - festival- & fasting-aware metabolic digital twin for Indians with type 2 diabetes

Uncertainty-aware, Physiology-informed Virtual Avatar for Adverse-event Surveillance · static EHR sets the prior, live wearables update the posterior, the festival calendar tells the twin what is coming

1 DATA STREAMS

Static / historical (EHR)

Synthea-India-lite generator
demographics, state, urban/rural, ABHA
HbA1c, FPG, lipids, TyG, eGFR, ALT, Lp(a)
diagnoses: T2D, HTN, neuropathy
meds: metformin, SU dose, basal insulin
genetics: CYP2C9, TCF7L2, PRS
vrat questionnaire: tradition, style,
SU habit when fasting

Dynamic / real-time (IoT)

CGM 5-min (Libre/Dexcom-like, lag+noise)
or 2 fingersticks/day (frugal tier)
smartwatch: HR, HRV (RMSSD), steps
sleep stages & efficiency
meal photo log (carbs, GI), smart pillbox
phone step counter (minimal tier)

Context (India-specific)

Festival & fasting calendar engine
Navratri, Ekadashi, Karva Chauth, Ramadan,
Diwali/Eid feasts, garba nights
CPCB-style PM2.5 (+0.40 mg/dL per 10 ug/m3)
Indian food GI library

2 CONSENT & INTEROP

ABDM-ready ingestion

Twin registers as Health Information User
consent artefact via HIE-CM (purpose CAREMGT)
FHIR R4 bundles (NRCeS IG): Patient,
Observation, MedicationStatement, Consent
DPDP Act 2023 / Rules 2025:
notice + consent, purpose limitation,
erase on consent expiry, breach < 72 h
Research/training: synthetic / anonymised
data only (Rule 16 research carve-out)

Feature fusion

per patient, every 30 min:
glucose trend (CGM or fingerstick)
wearables (HR dev, HRV ratio, steps, sleep)
meal / SU on board
calendar: fast type, minutes into fast,
minutes to fast-break, planned carbs,
planned garba
EHR + pharmacogenomics
+ 22 twin features (from layer 3)

3 DIGITAL TWIN CORE

EHR -> physiology prior

ridge maps labs+genes -> SI, beta-gain,
hepatic set-point, SU effect, SU half-life
(CYP2C9 & eGFR lengthen SU t1/2 prior)

SU-Bergman-Gut model (what the TWIN believes)

Bergman minimal model (G, X)
+ 2-compartment gut, GI-dependent k_abs
+ glucose-stimulated insulin secretion
+ sulfonylurea PK/PD (secretes at low G)
+ fasting clock: hepatic output falls >12 h
+ dawn, sleep->SI, activity, PM2.5,
counter-regulation blunted by neuropathy

Particle-filter data assimilation

64 candidate physiologies per patient
predict -> update on each glucose reading
unannounced-meal particles
2-h calendar-conditioned forecast ensemble
self-check: normalised innovation (NIS)
-> drift flag -> recalibration

4 INTELLIGENCE

Hybrid risk model

gradient boosting on fused features
targets: hypo <70 & hyper >250 mg/dL
sustained >=15 min within next 120 min
isotonic calibration
alarm threshold for 80% recall
ablations: no-twin, no-calendar, rule

Uncertainty & explanation

Mondrian (class-conditional) conformal:
a margin per predicted-glucose band, because
one margin is 90% on average and ~50% in
the range a clinician is watching
counterfactual driver attribution
patient-clustered bootstrap CIs on every number

Vrat Rehearsal (what-if)

36-h replay on posterior particles
fast type x SU timing x low-GI swap
x walk x garba x basal insulin
-> P(hypo), P(severe), P(hyper), nadir

5 WHO USES IT

Clinician console

roster ranked by 2-h risk
CGM + twin forecast + conformal band
risk cards with drivers
virtual-patient physiology vs EHR prior
Vrat Rehearsal sandbox
twin fidelity chart, FHIR export

ASHA 'Vrat Saathi' list

pre-festival rehearsal for every faster
ranked by predicted hypo risk
language, ward, action script
-> eSanjeevani teleconsult

Back into the health system

FHIR RiskAssessment to HIP / EMR
NP-NCD / Ayushman Arogya Mandir
model card: degradation by sensor tier,
fairness by sex/region/age/eGFR

HOW HARD THE TEST IS

The virtual body runs a DIFFERENT model from the twin - 3-compartment gut, interstitial insulin delay, circadian insulin sensitivity, renal excretion, incretin - so error includes model misspecification, not just parameter error · Eating is messy on purpose, so the festival calendar is a PRIOR and not an oracle · Reported beside a 4-feature timetable baseline with no glucose and no twin · Patient-grouped CV, bootstrap CIs, leakage test

GOVERNANCE RAIL

CDSCO draft medical-device-software guidance (Oct 2025): informs, does not drive -> likely Class B · NIS drift monitor feeds a pre-declared Algorithm Change Protocol · ICMR 2023 AI ethics: human in the loop, Indian-representative data, non-discrimination audit, right to refuse AI · DPDP: consent-tagged data, purpose limitation, deletion on expiry · Sensor-tier degradation curve published so low-resource users are not silently under-served